ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 2 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Robotics paper 2 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, sensible design choices, equivalent code and neat, recognisable diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 A ✓ 1.1.3 C ✓ 1.1.4 D ✓ 1.1.5 B ✓ 1.1.6 A ✓ 1.1.7 C ✓ 1.1.8 C ✓ 1.1.9 B ✓ 1.1.10 A ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 False ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 F ✓ 1.3.3 J ✓ 1.3.4 H ✓ 1.3.5 D ✓ 1.3.6 A ✓ 1.3.7 E ✓ 1.3.8 G ✓ 1.3.9 I ✓ 1.3.10 B ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Battery with BMS and voltage regulator ✓; controller (e.g. Arduino / ESP32) in the centre ✓; any THREE sensors (ultrasonic, IR line sensors, wheel encoders, bump switch, PIR, battery-voltage sensing) ✓; motor driver (H-bridge) ✓ to two DC drive motors ✓; Wi-Fi module / router link to the supervisor's computer ✓ (arrows in the right direction). (6)

2.2 Differential drive ✓: it can turn on the spot in narrow aisles ✓ and is simple to build and control with two motors ✓ (Ackermann accepted only with a sound reason, e.g. better at high speed on long straight runs). (3)

2.3 Any FOUR clear, measurable specifications ✓✓✓✓, e.g. carries at least 5 kg; runs at least 1 hour per charge; top speed about 0,5 m/s; stops within 25 cm of an obstacle / person; fits through a 1 m aisle; reports battery level over Wi-Fi; has an emergency stop. (4)

2.4 $I = 2 \times 1{,}2 + 0{,}4 + 0{,}2 = 3\ A$ ✓; usable capacity $= 4\,000 \times 0{,}8 = 3\,200\ mAh = 3{,}2\ Ah$ ✓; $t = 3{,}2 \div 3$ ✓ $= 1{,}07\ h$ ✓ $\approx 1\ h\ 4\ min$, so it meets the one-hour requirement (only just) ✓ (5)

[18]

QUESTION 3

3.1 $v = \pi dN \div 60$ ✓ $= \pi \times 0{,}08 \times 120 \div 60$ ✓ $= 0{,}50\ m/s$ ✓ (3)

3.2 $t = s \div v$ ✓ $= 30 \div 0{,}503 = 59{,}68\ s$ ✓ (2)

3.3.1 $\text{Ratio} = 6\,000 \div 120$ ✓ $= 50:1$ ✓ (2)

3.3.2 $T = 0{,}02 \times 50$ ✓ $= 1\ \text{N·m}$ ✓ (2)

3.4 To turn left on the spot the left wheel turns backwards and the right wheel forwards at the same speed ✓✓. The encoders count each wheel's pulses; the controller compares them and speeds up / slows down one motor (PWM) so both wheels turn the same distance ✓✓. (4)

3.5 $a = (v - u) \div t = (0{,}5 - 0) \div 2 = 0{,}25\ \text{m/s}^2$ ✓; $F = m \times a$ ✓ $= 12 \times 0{,}25 = 3\ \text{N}$ ✓ (3)

[16]

QUESTION 4

4.1 Start terminal ✓; process "drive forward" ✓; input / process "measure distance" ✓; decision "distance < 25 cm?" ✓ (No → back to drive forward ✓); Yes → "stop" and "wait 3 s" ✓; decision "still blocked?" ✓; Yes → "turn right 90°" then back to drive forward, No → back to drive forward ✓ (correct symbols and arrows). (8)

4.2.1 The time (in microseconds) that the echo pin stays HIGH, i.e. the time for the sound to travel to the object and back ✓ (1)

4.2.2 The sound travels to the obstacle and back, so the distance is half of the total ✓ (1)

4.2.3 The speed of sound in cm per microsecond (343 m/s = 0,0343 cm/µs) ✓ (1)

4.2.4 Function header ✓; stopMotors() ✓; delay(3000) ✓; measure again ✓; if closer than 25 cm ✓ call turnRight90() ✓, e.g. (6)

```
void stopAndCheck() {
  stopMotors();
  delay(3000);
  if (readCm() < 25) {
    turnRight90();
  }
}
```

4.3 delay() freezes the program, so the robot cannot read its sensors or react (e.g. to a person stepping in front) while it waits ✓. With millis() the program keeps looping and checks whether enough time has passed ✓; a state machine keeps track of the current mode (driving, waiting, turning) so the robot can do several jobs at once and react quickly ✓. (3)

[20]

QUESTION 5

5.1 Any FOUR parts ✓✓✓: battery-voltage sensor / encoders (data source); microcontroller with Wi-Fi (e.g. ESP32); Wi-Fi access point / router / gateway; cloud or local server / IoT platform; dashboard app on the computer. Protocol ✓: MQTT (or HTTP / HTTPS, WebSocket). (3 marks for any four parts, 1 for the protocol) (4)

5.2 In order ✓✓✓✓✓: measure the sensor and design the bracket in CAD; export it as an STL file; slice it (orientation, layer height, infill, supports) to make G-code; print it; remove supports / clean up and test-fit the sensor. (5)

5.3 PETG ✓ – tougher, less brittle and handles heat and knocks better than PLA in a busy warehouse ✓ (PLA accepted with a sensible reason, e.g. cheap, easy to print and the bracket carries little load). (2)

5.4 Any TWO hazards ✓✓ each with a control ✓✓: collision with people – ultrasonic / PIR sensors, low speed, warning lights and beeper; tripping over the robot – bright colours, marked robot lanes; crushing / pinching at moving parts – covers and guards; battery fire – BMS, approved charger, charge in a safe area; falling parcels – load restraints, weight limit. (4)

5.5 In a clearly visible place that is easy to reach from any side of the robot (e.g. on top) ✓ (1)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE